

## **Experiment: Student Performance Prediction using Multiple Linear Regression**

### **Aim**

To develop a Student Performance Prediction System using the Multiple Linear Regression algorithm

### **Tools Required**

- Python
- Pandas
- Scikit-learn
- VS Code

### **Code:**

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = pd.read_csv("E:/New folder/New folder/Student_performance.csv")
X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Marks:", y_pred)
print("R2 Score:", r2_score(y_test, y_pred))
```

### **Output**

```
0.35889305 0.36870912 0.35303684 0.41602672 0.35534808 0.40327175
0.3815379 0.37049355 0.39295323 0.34497302 0.38532715 0.37349519
0.35510434 0.34550029 0.41942107 0.36159736 0.44113048 0.35761981
0.37355422 0.37855138 0.399507 0.35579281 0.34441043 0.39169336
0.41988877 0.35511014 0.43696117 0.40638599 0.41316148 0.40964936
0.37398669 0.37165046 0.4158893 0.38791832 0.41273133 0.36829711
0.37943036 0.37549944 0.34918078 0.36340089 0.40582018 0.43799898
0.39781757 0.36265492 0.44931211 0.33771118 0.41956096 0.34859143
0.34586847 0.37646847 0.34493845 0.37314244 0.32437468 0.41831238
0.40068816 0.42970029 0.39347971 0.40408799 0.376409 0.38431845
0.39950986 0.34902682 0.3938403 0.37823416 0.37439288 0.31293388
0.41589985 0.33676737 0.3674135 0.42783345 0.36568572 0.42713216
0.33589835 0.37863382 0.39029655 0.32652036 0.31455119]
R2 Score: -0.014649807606850196
PS C:\Users\chandravardhan reddy\AppData\Local\Programs\Microsoft VS Code> █
```

## Result

Multiple Linear Regression model was successfully implemented.